

DSA in Python

Complete Learning Roadmap — Hinglish Guide

Data Structure

Data ko kaise STORE karein

+

Algorithm

Us data pe kaam kaise karein

=

DSA

Dono milke kaam karte hain!

■ Step-by-Step Learning Path

1

STEP 1 — Python Basics Pakka Karo

2–3 Weeks

- Lists, Tuples, Dict, Sets
- OOP (Classes & Objects)
- Recursion Basic idea
- Functions & Loops
- Big O Notation (Basic)
- String Methods

2

STEP 2 — Basic Data Structures

4–5 Weeks

- Array / List
- Stack
- HashMap / Dictionary
- String
- Queue
- Linked List

3

STEP 3 — Basic Algorithms

3–4 Weeks

- Binary Search
- Merge Sort
- Two Pointer Technique
- Bubble / Selection Sort
- Quick Sort
- Sliding Window

4

STEP 4 — Advanced Data Structures

4–5 Weeks

- Binary Tree & BST
- Graph (BFS & DFS)
- Segment Tree
- Heap / Priority Queue
- Trie
- Disjoint Set

5

STEP 5 — Advanced Algorithms

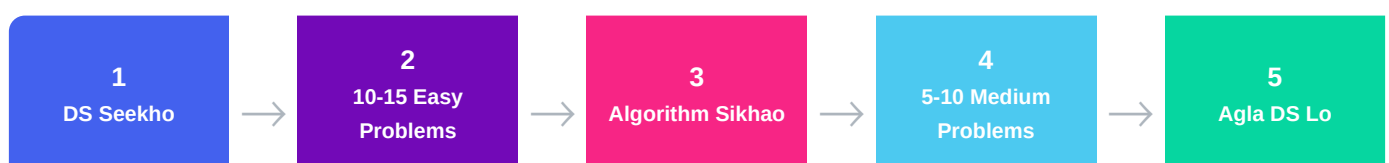
5–6 Weeks

- Recursion + Backtracking
- Greedy Algorithms
- Graph Algorithms
- Dynamic Programming (DP)
- Divide & Conquer
- Bit Manipulation

■ Data Structure ke Saath Algorithm — Order Table

#	Data Structure	Algorithm (Saath mein seekho)	Real-life Example
1	Array / List	Searching, Sorting, Two Pointer	Mobile contacts list
2	String	Pattern Matching, Anagram Check	Google search suggestions
3	Stack	Bracket Matching, Undo Feature	Browser back button
4	Queue	BFS, Task Scheduling	Ticket booking system
5	HashMap / Dict	Frequency Count, Two-Sum	Word count in book
6	Linked List	Traversal, Reverse, Cycle Detection	Music playlist
7	Binary Tree	Inorder / Preorder / Postorder	File system folders
8	BST	Insert, Delete, Search	Dictionary lookup
9	Heap	Top K Elements, Priority Queue	Hospital emergency queue
10	Graph	BFS, DFS, Dijkstra, Topological Sort	Google Maps navigation

■ Sabse Zaroori Formula — The Golden Loop



■ Pro Tips

- Roz minimum 1 problem zaroor solve karo
- Pehle apna solution khud socho, phir dekho
- Brute force se shuru karo, optimize baad mein
- Time complexity samjho har solution ka
- Code likhne se pehle pen-paper pe socho
- LeetCode Easy → Medium → Hard order follow karo
- Mistakes se seekho, editorial padho

■ Free Resources

- LeetCode.com — Problems practice
- GeeksForGeeks.org — Theory + Code
- visualgo.net — Animation se samjho
- YouTube: Abdul Bari (Algorithms)
- YouTube: CodeWithHarry (Python DSA)
- YouTube: NeetCode (LeetCode solutions)
- Book: Cracking the Coding Interview

■ 6-Month Realistic Timeline

Month 1	Python OOP + Big O + Arrays + Strings
Month 2	Stack, Queue, HashMap, Linked List
Month 3	Sorting Algorithms + Binary Search + Trees
Month 4	BST + Heap + Graph (BFS/DFS)
Month 5	Recursion + Backtracking + Dynamic Programming
Month 6	Greedy + Mixed Practice + Mock Interviews

■ Ek Line Summary

**"Pehle DS seekh → Turant uske problems solve kar → Phir Algorithm seekh → Problems solve kar →
Agle DS pe ja"**

Yahi ek loop hai jo tujhe DSA master bana dega! ■